Name: _____

Class: _____Date:_____

LESSON 6. POWERS WITH NATURAL EXPONENTS

1.51. Write each product in a shorter way using powers:

a) $3 \cdot 3 \cdot 3 \cdot 3$; b) $3 \cdot 4 \cdot 12 \cdot 12$; c) $3 \cdot 3 \cdot 7 \cdot 7 \cdot 7$.

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1.52. a) Complete the table of values of 3^n:

n	0	1	2	3	4	5	6	7
3^n								

b) Write the following numbers as powers of 3: 27; 243; 729; 2 187.

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1.53. a) Write the squares of the first eighteen natural numbers in order from smallest to largest;

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b) Write each of the following numbers as the square of a natural number: 81; 144; 225; 256; 324; 361.

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1.54. a) Calculate mentally 10^n for $n \in \{0; 1; 2; 3; 4; 5; 6\}$. State the general rule for calculating a power of 10;

b) Write the following numbers as powers of 10: 100; 1 000; 1 000 000; 100 000 000; 10 000 000 000.

1.55. Calculate:

a) 3^4; b) 4^3; c) $2^3 \cdot 3^2 \cdot 5$.

1.56. Find n, given:

a) $3^5 = n$; b) $n^3 = 216$; c) $11^n = 14\,641$.

1.57. Write the result of each calculation as a single power:

a) $5 \cdot 5^3 \cdot 5^6$; b) $6^5 : 6^3 : 6$; c) $(y^3)^5$.

1.58. Is the following statement true or false?

"No square number has 7 as its units digit."

1.59. Find the last digit of 57^5 and show that $57^5 + 2\,031^6$ is not a square number.

1.60. Compare without calculating the powers:

a) 8^7 and 16^5; b) 243^4 and 27^7; c) 3^{39} and 5^{26}.

1.61. Explain why the following three numbers are all square numbers:

a) A = 1 111 − 22; b) B = 111 111 − 222; c) C = 11 111 111 − 2 222.